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Complete Trigonometry Crash Course

Contents

I The Basic Tools **3**

1 Geometry Tools Needed to Study Trigonometry **4**
1.1 Right Triangles 4
1.1.1 The Pythagorean Theorem 5
1.1.2 Special Right Triangles 5
1.1.3 Classifying Triangles 7
1.2 Circles, Arcs, and Chords 9
1.2.1 Chords and Arcs 9
1.2.2 Area of Circles 11

2 Algebra Review **13**
2.1 Cartesian Coordinates 13
2.1.1 Writing Coordinates 14

Part I

The Basic Tools

Chapter 1

Geometry Tools Needed to Study Trigonometry

1.1 Right Triangles

There are three main types of angles:

- Acute Angles
- Right Angles
- Obtuse Angles

Definition 1 (Angles).
An **acute angle** is an angle with a measure less than 90°.
An **right angle** is an angle with a measure of 90°.
An **obtuse angle** is an angle with a measure greater than 90°.

Recall from geometry that you need three known measures to describe a triangle – the lengths of two sides and the measure of an angle, the lengths of three sides, or three other measures. For a right triangle, you need only two additional measures, because you already know that one of the angles' measure is 90°.

The following four cases describe all possible ways to describe a right triangle:

1. The lengths of two legs
2. The lengths of one leg and the hypotenuse
3. The lengths of one leg and the measure of an acute angle
4. The length of the hypotenuse and the measure of one acute angle

Suppose you need to calculate the lengths of all three sides. For cases 1 and 2, you only need geometry, specifically, the Pythagorean Theorem. But for cases 3 and 4, you need trig to get the remaining sides.

1.1.1 The Pythagorean Theorem

Let's take a look at the famous geometry tool *Pythagorean Theorem*, which helps us solve for cases 1 and 2.

Theorem 1 (Pythagorean Theorem)**.** The **Pythagorean Theorem** states that if a and b are the lengths of the legs of a right triangle and c is the length of the hypotenuse, then:

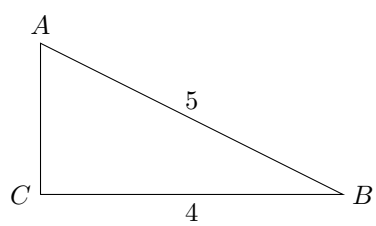
$$a^2 + b^2 = c^2$$

.

Let's illustrate how cases 1 and 2 can be solved with the help of the Pythagorean Theorem.

Sample Problem 1

Two legs of a right triangle measure 5 and 12 inches. Find the hypotenuse.



Since you know the length of the two legs, you can substitute those into the theorem's equation:

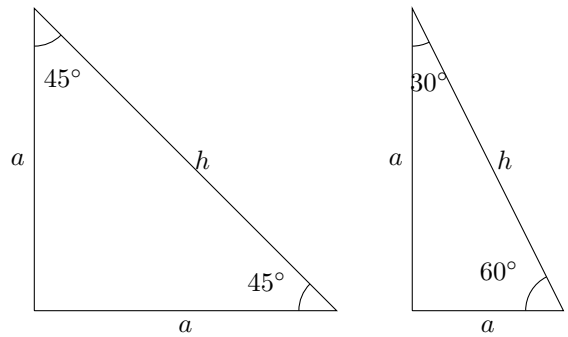
$$\begin{aligned} a^2 + b^2 &= c^2 \\ 5^2 + 12^2 &= c^2 \\ 25 + 144 &= c^2 \\ 169 &= c^2 \\ \sqrt{169} &= c \\ c &= 13 \end{aligned}$$

.

Solution: The length of the hypotenuse is 13 inches long.

1.1.2 Special Right Triangles

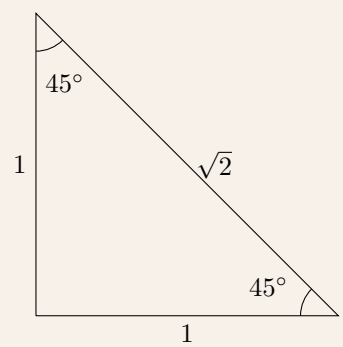
In this section, we'll take a look at two special kinds of right triangles. For these special triangles, if you know the length of any one side of the triangle, you can find the length of the other two sides.



The triangle on the left is called a $45° - 45° - 90°$ triangle, or an isosceles right triangle. The triangle on the right is called a $30° - 60° - 90°$ triangle. The Pythagorean Theorem is used to prove the following relationship that exist in these two special right triangles.

Definition 2 (Isosceles Right Triangle)**.** An **isosceles right triangle** is a triangle that has two equal sides and two equal angles, as well as a third angle that is a right angle. The only triangle that satisfies all these conditions is a $45° - 45° - 90°$ triangle.

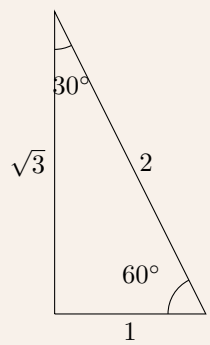
Theorem 2. If each acute angle of a right triangle measures $45°$, the hypotenuse is $\sqrt{2}$ times as long as a leg.



Example. If one leg in a $45° - 45° - 90°$ triangle is 10 units long, then the hypotenuse is $10\sqrt{2}$ units long.

Theorem 3. If acute angles of a right triangle measure $30°$ and $60°$, then the hypotenuse is twice as long as the shortest leg and the longer leg is $\sqrt{3}$

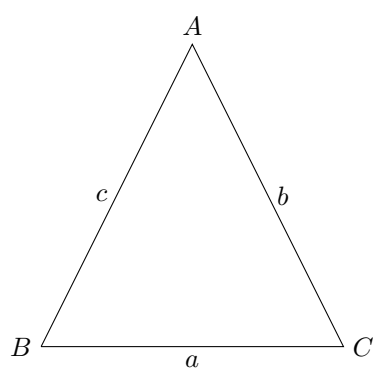
times as long as the shortest leg.



Example. If the shortest leg in a $30° - 60° - 90°$ triangle is 6 units long, then the hypotenuse is $2 \times 6 = 12$ units long, and the longest leg is $6\sqrt{3}$ units long.

1.1.3 Classifying Triangles

Any triangle is six elements: three sides and three angles. Throughout this book, I use capital letters (A, B, C) to denote measures of angles and lowercase letters. I also use lowercase letters (a, b, c) to denote sides of the triangles I use the small letter corresponding to the name of the angle opposite to this side. I also use the symbol $\triangle$ to identity a triangle. Therefore, an expression $\triangle ABC$ means "the triangle ABC".



When given the lengths of three sides, you can construct a triangle that is determined by these sides; therefore, its angles are determined as well. However, you need to be careful thinking this way, because not all arbitrary side lengths enable us to construct a triangle. Before constructing a triangle given its sides, you need to refer to the *Triangle Inequality*.

Definition 3 (Triangle Inequality)**.** The **triangle inequality** states that the sum of the lengths of any two sides of a triangle is greater than the length of the third side.

Example. Let's state whether it's possible for a triangle to exist with sides of the given lengths:

Sample 1: The given sides are 4, 5, and 6 units long. You need to compare the sums of any two sides with the third side. That means you need to exhaust all possibilities:

$$\begin{aligned} 4+5 &= 9 > 6 \\ 4+6 &= 10 > 5 \\ 5+6 &= 11 > 4 \end{aligned}$$
.

Sample 2: The given sides are 1, 2, and 3 units long. Let's try all possibilities:

$$\begin{aligned} 2+3 &= 5 > 1 \\ 3+1 &= 4 > 2 \\ 1+2 &= 3 = 3 \end{aligned}$$
.

The sum of these two sides (1 and 2) are not greater than the third side (3). Therefore, this triangle doesn't exist.

A triangle can be classified as acute (having three acute angles), right (having one right angle), and obtuse (having one obtuse angle). As we discussed, the lengths of the sides of a triangle determine its angles. Given the lengths of the sides, can you tell whether the triangle is acute, right, or obtuse?

The pythagorean theorem gives us a partial answer. We know that if the lengths satisfy the relationship $a^2 + b^2 = c^2$, then we know it's a right triangle. If it doesn't, then it's either an acute or obtuse triangle.

$$\begin{aligned} \text{If } \angle C \text{ of } \triangle ABC \text{ is acute, then } c^2 &< a^2 + b^2 \\ \text{If } \angle C \text{ of } \triangle ABC \text{ is obtuse, then } c^2 &> a^2 + b^2 \end{aligned}$$
.

Using this information and given the sides' lengths, you can identify the type of triangle you are dealing with.

Example. If the sides are 6, 7, 8, by substituting those values, you get:

$$8^2 = 64 < 6^2 + 7^2 = 36 + 49 = 85$$

.

So, this triangle is an acute triangle.

Note. In a triangle, the hypotenuse is always bigger in length than either of the sides. Hence, we always know which number should be substituted for c since it's the largest one of the three.

If the sides are 6, 8, and 10, then:

$$10^2 = 100 = 6^2 + 8^2 = 36 + 64 = 100$$

.

So the triangle is a right triangle.

If the sides are 5, 12, and 14, then:

$$14^2 = 196 > 5^2 + 12^2 = 25 + 144 = 169$$

.

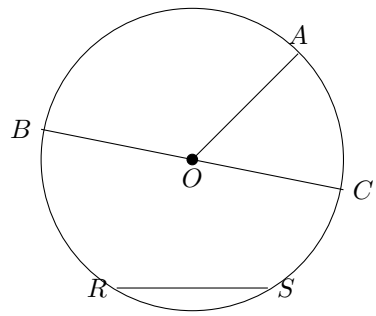
So the triangle is an obtuse triangle.

1.2 Circles, Arcs, and Chords

Another basic shape talked about in geometry is a circle. Let's review some basic elements of a circle because you use them a lot throughout trigonometry.

1.2.1 Chords and Arcs

Let's analyze the circle below and take note of its key elements. $\overline{OA}$ is a radius. $\overline{RS}$ is a chord. $\overline{BC}$ is a diameter.

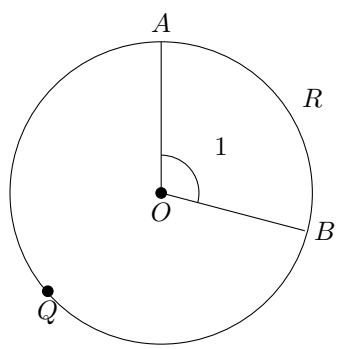


Definition 4 (Radius)**.** A **radius** of a circle is a segment that joins the center of a circle to a point on the circle.

Definition 5 (Chord)**.** A **chord** is a straight line segment joining two points that lie on the circumference of a circle. A chord whose endpoints lie opposite to each other on the circle and whose center passes through the circle's center is referred to as the diameter.

Definition 6 (Diameter)**.** A **diameter** is a chord that contains the center.

Now, let's discuss central angles and arcs of a circle.



$\angle 1$ is the central angle of the circle O. R is a minor arc between the points A and B, which is denoted as arc AB. The part that isn't between points A and B are referred to as the major arc AQB.

Note. To name a major arc, three letters must be used.

The measure of a minor arc is defined to be the measure of its central angle. If the measure of $\angle 1 = 40°$, you can write the measure of $AB = 40°$ as well. The measure of a major arc is calculated by subtracting $40°$ from $360°$:

$$360° - 40° = 320°$$

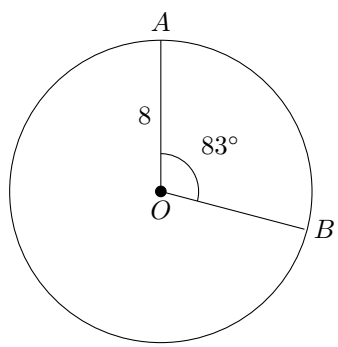
.

Definition 7 (Central Angle)**.** A **central angle** of a circle is an angle with a vertex at the center of the circle.

Definition 8 (Minor Arcj)**.** A **minor arc** of a circle is the union of two points on the circle and all the points of the circle that lie on the interior of the central angle with sides that contain the two points. Minor arcs measure less than 180°.

Definition 9 (Major Arc)**.** A **major arc** is an arc of a circle having measure greater than or equal to 180°.

Example.



In a circle of radius 8, how long is the chord of an arc of 83°? It turns out again that geometry doesn't have the tools to solve this problem. This problem provides a glimpse of types of problems that trigonometry will help solve.

1.2.2 Area of Circles

To find the area of a circle, you need to know its radius. You can then use the following formula to calculate its area.

$$A = \pi r^2$$

.

Example. If the radius is 7 inches, the area of the circle can be calculated by substituting this value into the formula:

$$A = \pi r^2 \Rightarrow \pi 7^2 = 49\pi$$

.

Chapter 2

Algebra Review

Basic algebra and trigonometry deal with two different areas of mathematics. Many people believe that trigonometry is a completely different discipline than algebra. But the truth is that isn't really the truth. You cannot fully comprehend trigonometry without knowing the basis of algebra, making algebra a prerequisite to trigonometry.

Algebra deals with finding the unknown variables and understanding functions – whereas trigonometry explores aspects of triangles, which are their sides and angles and finding a correlation between them. Algebra seeks to solve equations composed of multiple terms and to find their roots, whereas trigonometry focuses mainly on sin, cos, tan, degrees, radians, and polar coordinates.

It's helpful to imagine trigonometry as an extension to algebra because we often treat trigonometric concepts in algebraic terms. For example, solving a trigonometric equation is similar to solving an algebraic equation. You also learn the basic rules of vital topics like polynomials, exponents, graphing and much more in algebra. However, despite their similarities, there are still a number of differences. They are still both prerequisites to more advanced mathematical topics like calculus and differential equations.

2.1 Cartesian Coordinates

In algebra, you use the standard Cartesian coordinate system when graphing various functions. One of the most common methods of solving systems of equations is graphing them, where the solution is the point of intersection in all the lines.

Definition 10 (Cartesian Coordinate System)**.** The **cartesian coordinate system** specifies each point on a plane by a pair of numerical coordinates, which are distances from the point to two fixed perpendicular lines, commonly referred to as the x-axis and y-axis.

The simplest coordinate system is a number line on which you represent numbers. Next, we represent ordered pairs of numbers in a plane on which two of these lines are located.

In plane geometry, there are two axes at right angles that are usually called x-axis and y-axis. The position of any point in the plane can be given by its two coordinates (x, y). These coordinates give the point's distance in the x and

y directions from the origin, which is the point of intersection of the two axes. The origin is labeled with the pair of numbers both at zero – $(0, 0)$.

2.1.1 Writing Coordinates